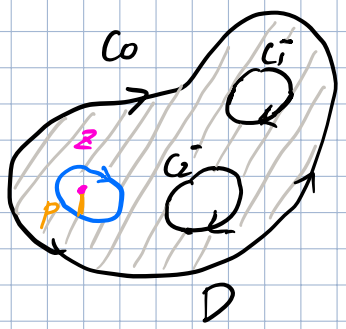
关键：把C积分化为 $\forall z$ 点充分小邻域边界积分 s.t. $f(\xi)$ 在那里几乎就是个常数.

然后估计 $\left| f(z) - \oint_{\partial N(z,\rho)} \frac{f(\xi)}{\xi - z} d\xi \right|$




证明：

① 在 $z$ 附近作小圆 $\partial N(z,\rho)$，于是 $f$ 在 $C_0 + C_1^- + C_2^- + \cdots + C_n^- + \partial N(z,\rho)^-$ 所围区域及边界上解析.

由 Cauchy 积分定理：

$$\left( \underbrace{\oint_{C_0 - C_1 - C_2 - \cdots}}_{C} + \oint_{-\partial N(z,\rho)} \right) \frac{f(\xi)}{z - \xi_0} d\xi = 0.$$

$$\Rightarrow \oint_C \frac{f(\xi)}{z-\xi} d\xi = \oint_{\partial N(z,\rho)} \frac{f(\xi)}{z-\xi} d\xi$$

② 比较 $\oint_{\partial N(z,\rho)} \frac{f(\xi)}{z-\xi} d\xi$ 与 $2\pi i \oint_{\partial N(z,\rho)} \frac{f(z)}{z-\xi} d\xi$

$$\left| \oint_{\partial N(z,\rho)} \frac{f(\xi) - f(z)}{\xi - z} d\xi \right| < \oint_{\partial N(z,\rho)} \frac{|f(\xi) - f(z)|}{|\xi - z|} |d\xi|$$

$$< \frac{\varepsilon}{\rho} \oint_{\partial N(z,\rho)} |Len\,\xi| = \frac{\varepsilon}{\rho} \cdot 2\pi\rho = \varepsilon.$$